Test Booklet

The 2026 Raleigh Division B Invitational, Presented by NCBioNetwork

Botany

DO NOT WRITE ON THIS TEST
Open this booklet when the event leader starts time
Tiebreakers: [7, 17, 18, 47, 52]

1. Which symptom is not observed with a plant experiencing a nutrient deficiency?
    a. Chlorosis
    b. Necrosis
    c. Turgid appearance
    d. Distorted Leaves

2. Which plant disease is characterized by sunken, oozing spots on stems, most common on stone fruits?
    a. Bacterial canker
    b. Fire blight
    c. Powdery mildew
    d. Root rot

3. What is not a treatment solution for powdery mildew?
    a. Baking soda spray
    b. Commercial spray
    c. Neem oil spray
    d. Salt water spray

4. Which plant disease is fungal?
    a. Anthracnose
    b. Crown gall
    c. Fire blight
    d. Black rot

5. Which crop is grown and harvested primarily for its oil?
    a. Rice
    b. Almonds
    c. Soybeans
    d. What

6. Farming, focusing on the cultivation of crops, is known as:
    a. Arable farming
    b. Pastoral farming
    c. Poultry farming
    d. Dairy farming

7. Cattle raised in grass-fed pastures may develop a disease caused by low-magnesium, high-potassium grasses. Which disease would this most likely be?
    a. Grass tetany
    b. Fescue toxicosis
    c. Nitrate poisoning
    d. Prussic acid poisoning

8. Why might a scientist genetically modify a crop?
    a. To increase drought susceptibility
    b. To increase yield
    c. To decrease pest resistance
    d. To decrease herbicide resistance

9. Which crops are the most common genetically modified worldwide?
    a. Cotton
    b. Granola
    c. Corn
    d. Soybeans

10. Competition of plants that are of the same species is known as:
    a. Sporadic competition
    b. Interspecific competition
    c. Intraspecific competition
    d. Family competition

11. Who is known as the “father of botany”?
    a. Theophrastus
    b. Carl Linnaeus
    c. Agnes Arber
    d. Joseph Banks

12. What type of leaf margin is demonstrated in this image?



a. Entire
b. Serrate
c. Crenate
d. Sinuate

13. Inflorescence is a term to describe the arrangement of flowers on a stem or branch of a plant. What inflorescence is demonstrated in this image of Snapdragon?



a. Catkin
b. Spike
c. Corymb
d. Raceme

14. The image in the previous question shows a Snapdragon, typically grown as an annual. Which of the following is also an annual; consider that the flower is grown in zones 1-8?
a. Cone flowers
b. Lantana
c. Peonies
d. Daylilies

15. What is the name of the pigment in plant cells used to absorb light energy needed to convert carbon dioxide and water into sugars and oxygen?
    a. Chloroplast
    b. Chlorophyll
    c. Chloropicrin
    d. Chlorohydrin

16. Where inside the chloroplast does the Calvin Cycle occur?
    a. Thylakoid
    b. Inner membrane
    c. Outer membrane
    d. Stroma

17. What is the key difference between photosystem I and photosystem II?
    a. Photosystem II produces ATP and provides electrons for photosystem I
    b. Photosystem I produces ATP and provides electrons for photosystem II
    c. Photosystem II absorbs light at longer wavelengths than photosystem I
    d. Photosystem I absorbs light at shorter wavelengths than photosystem II

18. Some plants, such as the ghost pipe (Monotropa uniflora), do not perform photosynthesis. Which is not a way that plants that lack chlorophyll perform photosynthesis to obtain their energy?
    a. Haustoria
    b. Parasitism
    c. Crassulacean Acid Metabolism
    d. Mycoheterotrophy

19. What plant is a non-vascular bryophyte?
    a. Ferns
    b. Conifers
    c. Moss
    d. Horsetails

20. A structure in a plant transports sugars and other nutrients from source tissues, like the leaves, to sink tissues where they are needed for growth or storage. What is this structure?
    a. Xylem
    b. Collenchyma
    c. Sclerenchyma
    d. Phloem

21. Which of the following is the oldest living plant?
    a. Quaking aspen
    b. Seagrass
    c. Patagonia cypress
    d. Great Basin bristlecone pine

22. Paleobotanists study the fossils of plants. Which of the following is not something that a paleobotanist would study?
    a. Mineralized wood
    b. Leaf Imprints
    c. Pollen in amber
    d. Ancient bacteria

23. The study of pollen grains and other spores, especially as found in archaeological or geological deposits, is known as:
    a. Ichnology
    b. Palynology
    c. Paleoecology
    d. Herbology

24. A chemical process, particularly used on coal, to break down the material and reveal preserved plant tissues like spores, cuticles, and mycelia, is known as:
    a. Maceration
    b. Stratigraphy
    c. Palynology
    d. Radiometric dating

25. What are the four major classifications of plants?
    a. Bryophytes, pteridophytes, gymnosperms, angiosperms
    b. Bryophytes, pteridophytes, teratospermie, arbors
    c. Gymnosperms, angiosperms, conifers, vascular plants
    d. Gymnosperms, angiosperms, flowering plants, vascular plants

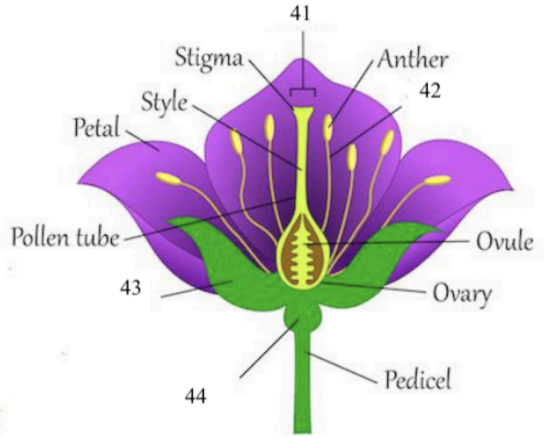
26. What classification does this plant most likely belong to?



    a. Gymnosperm
    b. Pteridosperms
    c. Ginkgoes
    d. None of the above

27. What is the main role of sclereids?
    a. Provide mechanical support and protection
    b. Serves as the immune system
    c. Helps ground the plant within its roots
    d. Help in the production of photosynthesis

Questions 28-30 utilize the image below:



28. What is the name of the structure labeled 41?
    a. Pistil
    b. Filament
    c. Sepal
    d. Receptacle

29. What is the name of the structure labeled 42?
    a. Pistil
    b. Filament
    c. Sepal
    d. Receptacle

30. What is the name of the structure labeled 43?
    a. Pistil
    b. Filament
    c. Sepal
    d. Receptacle

31. What is often associated with asexual reproduction utilized by plants?
    a. Pollination
    b. Germination
    c. Rhizomes
    d. Fertilization

32. The male sex organ of a plant is known as the:
    a. Stamen
    b. Pistil
    c. Stigma
    d. Style

33. A _____, also known as a runner, is a horizontal stem that grows above the ground, allowing a plant to spread and reproduce asexually? Fill in the blank.
    a. Stolon
    b. Bulb
    c. Rhizome
    d. Tuber

34. During sexual reproduction, the ovaries in the female reproductive organs mature into a fruit. What is the purpose of the fruit?
    a. Keep the seeds near the parent plant
    b. Serve as a food source for the plant
    c. Help attract pollinators for future reproduction
    d. Protect developing seeds

35. During sexual reproduction, the _____ produces the male gamete (found in pollen) and the female gamete (the egg inside the ovule) through mitosis. Fill in the blank.
    a. Diplophyte
    b. Halophyte
    c. Gametophyte
    d. Sporophyte

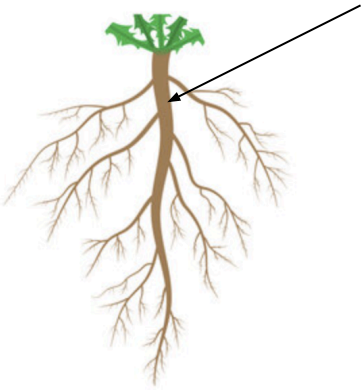
36. What type of stem is depicted in the image below?



    a. Herbaceous
    b. Woody
    c. Tendril
    d. Cladode

37. The total weight of organic matter in plants, including all stems, leaves, roots, flowers, and fruits that serve as a renewable source of energy and materials, is known as:
    a. Biomass
    b. Bacteria and fungi
    c. Solar energy
    d. None of the above

38. What type of root is the arrow pointing at in the image below?



    a. Fibrous root
    b. Adventitious root
    c. Taproot
    d. Prop root

39. A type of biological control that aims to increase the abundance and effectiveness of natural enemies (like predators, parasitoids, and microbes) that are already present in the local environment best describes:
    a. Classical control
    b. Augmentative control
    c. Conservation control
    d. Neo control

40. What is not a downside to biological control in plants?
    a. Typically slow results
    b. Unintended harm to non-target species
    c. Potential agent failure
    d. Possibility to avoid pest resistance

41. The transfer of pollen from the anther of one plant to the stigma of another plant is known as?
    a. Pollination
    b. Germination
    c. Fertilization
    d. Propagation

42. In a Punnett square of two heterozygous tall plants (Tt × Tt), what percent of the offspring are expected to be short (tt)?
    a. 75%
    b. 50%
    c. 25%
    d. 0%

43. The image below depicts a Longleaf Pine (*Pinus palustris*). Pine trees often have both male and female flowers. Which of the following best describes this characteristic?



    a. Dioecious
    b. Monoecious
    c. Diploid
    d. Haploid

44. Which of the following were the dominant plants of the Carboniferous Period?
    a. Flowering plants
    b. Gymnosperms
    c. Large, seedless vascular plants
    d. Algae

45. What is a defining characteristic of gymnosperms?
    a. Seeds are encapsulated in fruits
    b. Seeds are “naked” in cones
    c. They only produce spores
    d. They do not produce seeds

46. Which of the following is an evolutionary advantage of pollen?
    a. Tastes poorly to predators
    b. Naturally deters insects
    c. Allows for spore dispersal underwater
    d. Fertilization can occur without water

47. During what period did angiosperms become the dominant land plants?
    a. Devonian
    b. Triassic
    c. Carboniferous
    d. Cretaceous

48. Robert Hooke, a 17th-century English polymath, used the microscope to observe plant matter in cork tissue and coined which term?
    a. Thylakoid
    b. Spore
    c. Chloroplast
    d. Cell

49. Which Swedish scientist developed the modern system of naming plants, binomial nomenclature?
    a. Gregor Mendel
    b. Theophrastus
    c. Charles Darwin
    d. Carl Linnaeus

50. Binomial nomenclature gives every species:
    a. Two common names
    b. One genus and one species name
    c. Two latin names
    d. One genus, one species, and one common name

51. Select the answer choice that demonstrates the overall chemical equation for photosynthesis:
    a. $6\ CO_2 + 6\ H_2O \rightarrow C_6H_{12}O_6 + 6\ O_2$
    b. $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$
    c. $CO_2 + H_2O \rightarrow CH_4 + O_2$
    d. $6\ O_2 \rightarrow 6\ CO_2 + C_6H_{12}O_6$

52. Chlorophyll mainly absorbs which wavelengths of light?
    a. Red and blue
    b. Green and yellow
    c. Yellow and green
    d. Red and yellow

53. Why is photorespiration considered wasteful?
    a. It takes a long time to occur
    b. It produces excess oxygen
    c. It consumes oxygen and releases carbon dioxide without making sugar
    d. None of the above, it is not wasteful

54. What key role do plants play in the carbon cycle?
    a. Release of methane into the atmosphere
    b. Produce fossil fuels
    c. Break down dead plant matter
    d. Remove carbon dioxide from the atmosphere during photosynthesis

55. Cellular respiration is critical for plants by which it:
    a. Releases nitrogen gas, a waste product for the plant
    b. Converts chemical energy in sugars into usable ATP
    c. It produces chlorophyll that the plant utilizes for photosynthesis
    d. Glucose is generated for storage and future use

56. Both plants and nitrogen-fixing bacteria benefit from their interactions with one another. This would be an example of which relationship?
    a. Mutualism
    b. Commensalism
    c. Dualism
    d. Parasitism

57. The major reservoir of carbon in the biosphere is found in:
    a. Forest canopies
    b. The atmosphere
    c. Sedimentary rock
    d. Polar ice caps

58. Which is the primary way that deforestation disrupts the global carbon cycle?
    a. Increased photosynthesis produces more oxygen, altering carbon levels
    b. Increased photosynthesis increases nitrogen fixation, altering carbon levels
    c. Decreased photosynthesis results in lower ocean acidity
    d. Decreased photosynthesis results in higher carbon dioxide

59. How do plants mainly absorb phosphorus in the phosphorus cycle?
    a. From the air
    b. From water vapor
    c. From ions in the soil
    d. From nitrogen gas

60. Why can many animals not digest cellulose in plant material?
    a. The fat content is too high
    b. They lack enzymes to break glycosidic bonds
    c. The hydrogen bonds between amino acids are too strong
    d. They can, it is water-soluble and is quickly digested

61. Wood is primarily composed of
    a. Lignin and cellulose
    b. Pectin and chlorophyll
    c. Collagen and starch
    d. Proteins and keratin

62. Plants provide humans with structural materials because of:
    a. Elastic starch in xylem
    b. Cell walls made of cellulose and lignin
    c. Protein-based stems
    d. Collagen fibers in phloem

63. A competition strategy exhibited by plants by which they release chemicals into the soil to inhibit nearby competitors is known as:
    a. Allelopathy
    b. Chemical commensalism
    c. Autotoxicity
    d. Mycorrhizal signaling

64. Understory plants have unique adaptations to compete for resources with plants in the main canopy. Which is an example of these adaptations?
    a. High light compensation points
    b. Thick bark and reduced leaves
    c. Broad, thin leaves
    d. Photosynthesis occurs without chlorophyll

65. In a field, fast-growing weeds that colonize open soil before slower-growing perennials represents:
    a. Primary succession
    b. Secondary succession
    c. Tertiary succession
    d. None of the above

66. The first widely grown GMO crop in the United States was:
    a. Corn
    b. Soybeans
    c. Rice
    d. Apples

67. The enzyme used to “cut” DNA at specific sequences during genetic engineering is a:
    a. DNA polymerase
    b. Restriction enzyme
    c. DNA Ligase
    d. Reverse transcriptase

68. The plasmid in genetic modification is:
    a. A protein enzyme
    b. A storage vacuole in roots
    c. A type of plant pigment
    d. A circular piece of DNA

69. Which refers to the process of transferring beneficial genes using a bacterium such as *Agrobacterium tumefaciens*?
    a. Transcription
    b. Transformation
    c. Translation
    d. Transpiration

70. Fungal infections in plants often produce:
    a. Overproduction of chlorophyll
    b. Insufficient nitrogen fixation
    c. Leaf spots or moldy growths
    d. Smooth, shiny leaves

North Carolina State University Invitational - Repaired practice edition

Division B | Botany | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

4. (1 points) Authored replacement for source item 4: A supplied diagnostic report identifies Colletotrichum, a fungal genus, as the cause of a plant infection named anthracnose. Is this diagnosed infection fungal or bacterial? Explain using the report.

Answer: _____
_____
_____

9. (1 points) Authored replacement for source item 9: Supplied crop survey: soybeans 40 hectares, corn 30, cotton 20. Which of these genetically modified crop plantings occupies the largest area in this survey?

Answer: _____
_____
_____

14. (1 points) Authored replacement for source item 14: A plant germinates, flowers, sets seed and dies within one growing season. Classify its life cycle as annual, biennial or perennial.

Answer: _____
_____
_____

21. (1 points) Authored replacement for source item 21: Compare supplied ages of individual trees: A=600 years, B=1,200 years, C=4,000 years. Which individual is oldest? Explain why this comparison does not establish the age of a clonal colony.

Answer: _____
_____
_____

57. (1 points) Authored replacement for source item 57: The repair defines its reservoirs as sedimentary rocks, the atmosphere, living forest biomass and ice. Supplied carbon stores are 80, 10, 8 and 2 arbitrary units respectively. Which reservoir stores the most carbon in this model?

Answer: _____
_____
_____

60. (1 points) Authored replacement for source item 60: An animal can digest starch but cannot hydrolyze the beta-1,4 glycosidic links in cellulose because it lacks cellulase. Explain why “it cannot break any glycosidic bonds” is too broad.

Answer: _____
_____
_____

66. (1 points) Authored replacement for source item 66: A crop receives an inserted gene using a laboratory transformation method. Another crop is selected only through conventional crossing. Which is genetically engineered by the stated method?

Answer: _____
_____
_____